19 Given that $9^{-\frac{1}{2}} = 27^{\frac{1}{4}} \div 3^{x+1}$
find the exact value of x.

$9^{(-1/2)} = (3^2)^{(-1/2)} = 3^{-1}$.

$27^{(1/4)} = (3^3)^{(1/4)} = 3^{(3/4)}$.

$3^{-1} = 3^{(3/4)} \div 3^{(x+1)} = 3^{(-x-1/4)}$.

$-1 = -x - 1/4$, so $x = 3/4$.

3 marks total
Convert to base 3
Use laws of indices
x = ¾

$x =$ .....

(Total for Question 19 is 3 marks)